

(Marks)**Question 1**

(a) Evaluate $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$ (2)

(b) Expand $(x+3)^4$ (2)

(c) Differentiate $x^2 \cos^{-1} 3x$ (2)

(d) Find the remainder when the polynomial $P(x) = x^3 - 4x + 2$ is divided by $x + 4$ (1)

(e) Sketch the curve $y = x^3 - 4x$ and hence solve $x^3 - 4x \geq 0$ (2)

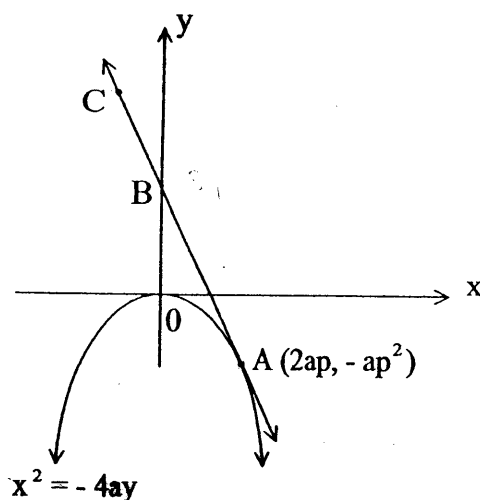
(e) Use the substitution $u = \log_e x$ to find the exact value of $\int_e^{e^2} \frac{1}{x \log_e x} dx$ (3)

(Marks)

Question 3**Start a new page**

(a) Find the term independent of x in the binomial expansion of $(2x - \frac{1}{x^2})^6$ (2)

(b) The point $A(2ap, -ap^2)$ is a variable point on the parabola $x^2 = -4ay$. The tangent at A meets the y axis at B . The point C lies on the tangent and divides AB externally in the ratio $3 : 1$



(i) Show that the equation of the tangent at A is $px + y = ap^2$ (2)

(ii) Find the coordinates of the points B and C (3)

(iii) Show that the locus of C is a parabola (1)

(c)

(i) Show that the function $f(x) = xe^x - 1$ has a zero between $x = 0$ and $x = 1$ (1)

(ii) Using $x = 0.5$ as a first approximation, use Newton's Method once to obtain a second approximation to the zero correct to 3 significant figures. (3)

(Marks)**Question 4****Start a new page**

(a) Solve $\sin 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$ (2)

(b) Find the exact value of

$$\int_0^{\frac{\pi}{12}} \sin^2 2x \, dx \quad (3)$$

(c) At any time t , the rate of cooling of the temperature T of a body is given by

$$\frac{dT}{dt} = -k(T - S)$$

where S is the temperature of the surroundings and k is a constant.

(i) Verify that $T = S + Ae^{-kt}$ is a solution of $\frac{dT}{dt} = -k(T - S)$ where A is a constant (1)

(ii) A cup of hot tea cooled from 100°C to 70°C in 10 minutes in a room where the temperature was 20°C . Find the values of A and k (2)

(d) Prove by mathematical induction that

$$11 \times 2! + 19 \times 3! + 29 \times 4! + \dots + (n^2 + 5n + 5)(n+1)! = (n+4)(n+2)! - 8$$

for $n = 1, 2, 3, \dots$

(4)

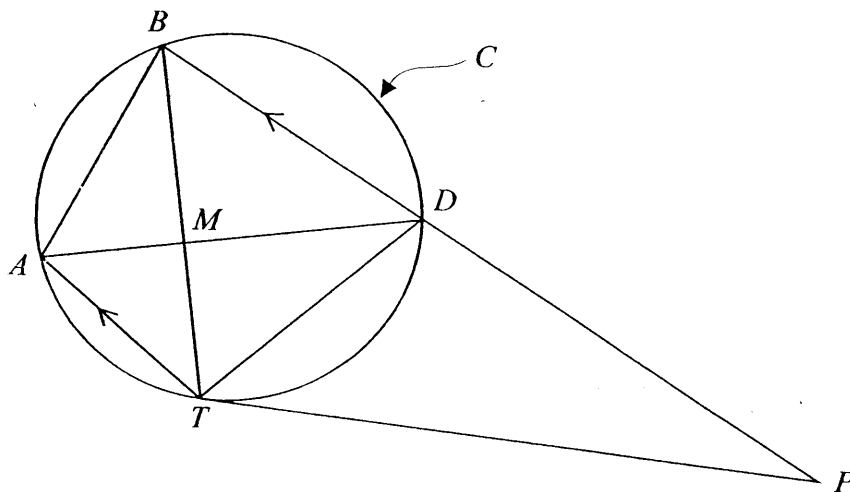
(Marks)

Question 5**Start a new page**

- (a) Find the acute angle between the lines $2x + y = 4$ and $x - y = 2$.

Answer to the nearest degree.

(2)



The diagram above shows a circle C . The points A , B , D and T lie on C and the point P is an exterior point.

PT is a tangent to C at T . The line AT is parallel to the line BP and point D lies on BP . The lines AD and BT intersect at M .

Copy or trace the diagram onto your page.

- (i) Prove that $\triangle PTB$ is similar to $\triangle BAT$ (3)

- (ii) Show that $AB = DT$ (2)

- (b) A particle is moving in a straight line with acceleration given by

$$\frac{d^2x}{dt^2} = 9(x - 2)$$

where x is the displacement in metres from an origin O after t seconds. Initially, the particle is 4 metres to the right of O so that $x = 4$ and has velocity $v = -6$

- (i) Show that $v^2 = 9(x - 2)^2$ (2)

- (ii) Find an expression for v and hence find x as a function of t . (3)

(Marks)

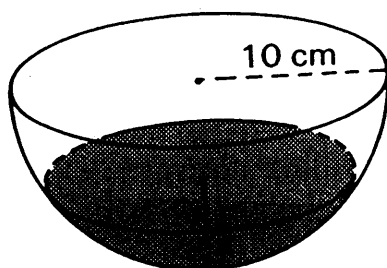
Question 6**Start a new page**

(a) The circle $x^2 + y^2 = 100$ is rotated about the y axis from $y = 10$ to $y = 10 - a$.

Use $V = \pi \int_{10-a}^{10} (100 - y^2) dy$ to show that the volume, V , is $V = \frac{\pi}{3} a^2 (30 - a)$

(3)

(b)



The volume of water in a hemispherical bowl of radius 10cm is given by $V = \frac{\pi}{3} x^2 (30 - x)$ where x cm is the depth of the water at any time t .

The bowl is being filled at a constant rate of $2\pi \text{ cm}^3 / \text{min}$

At what rate is the depth increasing when the depth is 2cm?

(3)

(c) A particle moves in a straight line and its position at time t seconds is given by

$$x = \sqrt{3} \sin \frac{t}{2} + \cos \frac{t}{2}$$

(i) Write $\sqrt{3} \sin \frac{t}{2} + \cos \frac{t}{2}$ in the form $R \sin (\frac{t}{2} + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ (2)

(ii) Hence prove that the particle is moving in simple harmonic motion about $x = 0$. (2)

(iii) For $0 < t < 4\pi$, when is the speed of the particle equal to 0.5 ms^{-1} . (2)

(Marks)

Question 7**Start a new page**

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 6x}{5x}$ (1)

(b) A football is kicked at an angle of α to the horizontal. The position of the ball at time t seconds is given by

$$x = vt \cos \alpha$$

$$y = vt \sin \alpha - \frac{1}{2}gt^2$$

where $g \text{ m/s}^2$ is the acceleration due to gravity and $v \text{ m/s}$ is the initial velocity of projection. (You are NOT required to derive these.)

(i) Show that the equation of the path of the ball is (2)

$$y = x \tan \alpha - \frac{gx^2}{2v^2} \sec^2 \alpha.$$

(ii) Show that the maximum height h reached is given by (3)

$$h = \frac{v^2 \sin^2 \alpha}{2g}$$

(iii) Hence show that $y = x \tan \alpha \left(1 - \frac{x \tan \alpha}{4h}\right)$. (2)

(iv) If $g = 10 \text{ m/s}^2$, $\alpha = 30^\circ$ and the ball just clears the head of a player 1.6m tall and 10m away, calculate the maximum height reached by the ball. (2)

(c) The real number x is a solution of the equation $x^2 - x - 1 = 0$. Use the Binomial Theorem to show that the sum S of the series $1 + x + x^2 + \dots + x^{2n-1}$ is given by

$$S = \sum_{r=1}^n {}^nC_r x^{r+1} \quad (2)$$

END OF PAPER

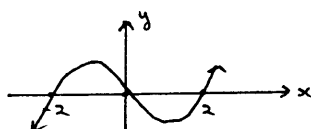
Question 1

$$\begin{aligned}
 \text{(a)} \quad \int_0^3 \frac{dx}{\sqrt{9-x^2}} &= \left[\sin^{-1} \frac{x}{3} \right]_0^3 \\
 &= \sin^{-1} 1 - \sin^{-1} 0 \\
 &= \frac{\pi}{2} - 0 \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad (x+3)^4 &= x^4 + 4x^3 \cdot 3 + 6x^2 \cdot 3^2 + 4x \cdot 3^3 + 3^4 \\
 &= x^4 + 12x^3 + 54x^2 + 108x + 81
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad y &= x^2 \cos^{-1} 3x \\
 y' &= uv' + vu' \\
 &= x^2 \cdot \frac{-1}{\sqrt{1-9x^2}} \cdot 3 + \cos^{-1} 3x \cdot (2x) \\
 &= \frac{-3x^2}{\sqrt{1-9x^2}} + 2x \cos^{-1} 3x
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad P(-4) &= (-4)^3 - 4(-4) + 2 \\
 &= -64 + 16 + 2 \\
 &= -46
 \end{aligned}$$

$$\begin{aligned}
 \text{(e)} \quad y &= x^3 - 4x \\
 &= x(x^2 - 4) \\
 &= x(x-2)(x+2)
 \end{aligned}$$


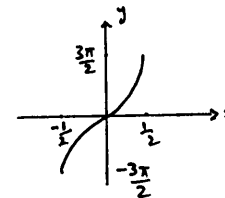
$$\begin{aligned}
 x=1, y &= 1^3 - 4(1) \\
 &= -3
 \end{aligned}$$

$$\begin{aligned}
 \therefore x^3 - 4x &\geq 0 \\
 -2 \leq x \leq 0, x &\geq 2
 \end{aligned}$$

$$\begin{aligned}
 \text{(f)} \quad x &= e^2, u = \log_e e^2 = 2 \\
 x &= e, u = \log_e e = 1 \\
 u &= \log_e x \\
 \frac{du}{dx} &= \frac{1}{x} \\
 \int_{e^2}^e \frac{1}{x \log_e x} dx &= \int_1^2 \frac{1}{u} du \\
 &= [\ln u]_1^2 \\
 &= \ln 2 - \ln 1 \\
 &= \ln 2
 \end{aligned}$$

Question 2

$$\begin{aligned}
 \text{(a)} \quad y &= 3 \sin^{-1} 2x \\
 -1 &\leq 2x \leq 1 \\
 -\frac{1}{2} &\leq x \leq \frac{1}{2}
 \end{aligned}$$

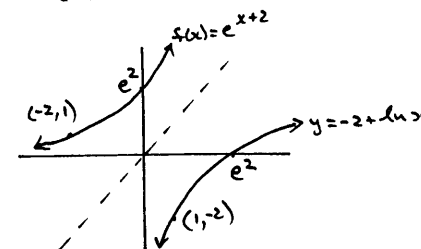


$$\begin{aligned}
 \text{(b)} \quad \angle DCB &= x \text{ (angles standing on the same arc are equal)} \\
 \angle DBC &= 90^\circ \text{ (angle in a semi-circle)} \\
 \angle BDC &= 65^\circ \text{ (alternate } \angle \text{ theorem)} \\
 x + 90 + 65 &= 180^\circ \text{ (angle sum of a } \Delta) \\
 x &= 25^\circ
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad P(x) &= x^3 - 2x^2 + kx + 24 \quad a=1, b=-2, c=k, d=24 \\
 \text{i)} \quad \alpha + \beta + \gamma &= -\frac{b}{a} \\
 &= 2 \\
 \text{ii)} \quad \alpha\beta\gamma &= -\frac{d}{a} \\
 &= -24 \\
 \text{iii)} \quad \text{let } \beta &= -\alpha \\
 \alpha - \alpha + \gamma &= 2 \\
 \gamma &= 2 \\
 P(2) &= 2^3 - 2(2)^2 + k(2) + 24 = 0 \\
 k &= -12
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad \text{i)} \quad f(x) &= e^{x+2} \\
 x &= e^{y+2} \\
 \ln x &= y+2 \\
 y &= -2 + \ln x \\
 f^{-1}(x) &= -2 + \ln x
 \end{aligned}$$

ii)



For full marks, intercepts to be shown.

Question 4

a) $\sin 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$

$$2\sin\theta \cos\theta - \sin\theta = 0$$

$$\sin \theta (2 \cos \theta - 1) = 0$$

$$\sin \theta = 0 \qquad \cos \theta = \frac{1}{2}$$

$$\theta = 0, \pi, 2\pi, \quad \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$b) \int_0^{\frac{\pi}{2}} \sin^2 2x \, dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 - \cos 4x \, dx \quad \begin{matrix} \sin^2 x = \frac{1}{2}(1 - \cos 2x) \\ \sin^2 2x = \frac{1}{2}(1 - \cos 4x) \end{matrix}$$

$$\left[\frac{x^2}{2} - \frac{1}{4} \sin 4x \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{12} - \frac{1}{4} \sin \frac{\pi}{3} \right) - \left(0 - \frac{1}{4} \sin 0 \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{12} - \frac{1}{4} \cdot \frac{\sqrt{3}}{2} - 0 \right]$$

$$= \frac{\pi}{24} - \frac{\sqrt{3}}{16}$$

c) i) $T = S + Ae^{-kt}$

2-9

$$\frac{dT}{dt} = -k \cdot A e^{-kt}$$

$$y - y_1 = m(x - x_1)$$

$$y + ap^2 = -p(x - 2ap)$$

$$y + ap^2 = -px + 2ap^2$$

$$px + y = ap^2$$

ii) $A=80, S=20, T=70, t=10$ $100 = 20 + Ae^0$
 $-10k$ $A=80$

$$C = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{3 \times 0 - 1 \times 2ap}{3-1}, \frac{3 \times ap^2 - 1 \times -ap^2}{3-1} \right)$$

$$= (-ap, 2ap^2)$$

iii) Let $n=1$, $LHS = 1 \times 2!$
 $= 22$

$$y = 2ax^{\frac{1}{2}}$$

$$x^2 = \frac{1}{2}ay \text{ which is a parabola}$$

$$\begin{aligned} \text{RHS} &= 5 \times 3! - 8 \\ &= 30 - 8 \\ &= 22 \\ &= \text{LHS} \end{aligned}$$

ii) $f'(x) = e^x + xe^x$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{0.5e^{0.5} - 1}{e^{0.5} + 0.5e^{0.5}} = 0.5711$$

$$\begin{aligned}
 \text{LHS} &= (k+4)(k+2)! - 8 + (k^2 + 2k + 1 + 5k + 10)(k+2)! \\
 &= (k+4)(k+2)! + (k^2 + 7k + 11)(k+2)! - 8 \\
 &= (k+2)! [(k+4) + (k^2 + 7k + 11)] - 8 \\
 &= (k+2)! (k^2 + 8k + 15) - 8 \\
 &= (k+2)! (k+5)(k+3) - 8 \\
 &= (k+3)! (k+5) - 8 \\
 &= \text{RHS}.
 \end{aligned}$$

Since the statement is true for $n=1$, it is true for $n=1+1=2$ & therefore true for all integers $n \geq 1$

QUESTION 5

$$\begin{aligned}
 a) \quad 2x + y &= 4 \\
 y &= -2x + 4 \quad m_1 = -2 \\
 x - y &= 2 \\
 y &= x - 2 \quad \therefore m_2 = 1
 \end{aligned}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-2 - 1}{1 + (-2)(1)} \right|$$

$$= \frac{3}{1} = 3$$

$$\therefore \theta = 72^\circ$$

- b) $\angle BTA = \angle TBP$ (alternate $\angle$ s, $BD \parallel AT$)
 $\angle BAT = \angle BTP$ (alternate segment theorem)
 $\therefore \triangle BAT \parallel \triangle TBP$ (corresponding angles equal)

ii) In $\triangle ABT$, $\triangle ADT$

AT is common

$\angle ABT = \angle ADT$ (angles on same arc AC are equal)

$\angle BAD = \angle BTD$ (angles on arc BD are equal)

$\angle TAD = \angle TDB$ (angles on arc TD are equal)

$\angle BTA = \angle ATD$ (alternate $\angle$'s $BD \parallel AT$)

$\angle BAT = \angle DTA$

$\therefore \triangle ABT \cong \triangle ADT$ (AAS)

$\therefore AB = DT$

Question 5 (continued)

$$i) \quad \frac{d^2 x}{dt^2} = 9(x-2)$$

$$\frac{1}{2} v^2 = \int 9(x-2)$$

$$\frac{1}{2} v^2 = 9 \left(\frac{x^2}{2} - 2x \right) + C$$

$$v = -6, x = 4, \quad \frac{1}{2} (-6)^2 = 9 \left(\frac{16}{2} - 8 \right) + C$$

$$C = 18$$

$$\frac{1}{2} v^2 = 9 \left(\frac{x^2}{2} - 2x \right) + 18$$

$$v^2 = 18 \left(\frac{x^2}{2} - 2x \right) + 36$$

$$= 9x^2 - 36x + 36$$

$$= 9(x^2 - 4x + 4)$$

$$= 9(x-2)^2$$

$$ii) \quad v = \pm 3(x-2)$$

but $v < 0$ when $x = 4$

$$\therefore v = -3(x-2)$$

$$\frac{dx}{dt} = -3(x-2)$$

$$\frac{dt}{dx} = \frac{1}{-3(x-2)}$$

$$t = -\frac{1}{3} \ln(x-2) + C$$

$$t = 0, x = 4, \quad 0 = -\frac{1}{3} \ln(4-2) + C$$

$$0 = -\frac{1}{3} \ln 2 + C$$

$$C = \frac{1}{3} \ln 2$$

$$\therefore t = -\frac{1}{3} \ln(x-2) + \frac{1}{3} \ln 2$$

$$t - \frac{1}{3} \ln 2 = -\frac{1}{3} \ln(x-2)$$

$$-3t + \ln 2 = \ln(x-2)$$

$$e^{\ln 2 - 3t} = x - 2$$

$$x = 2 + e^{\ln 2 - 3t}$$

! want if a
constant value
- when $t=0$
- $x=4$

Question 6

$$\begin{aligned}
 a) \quad V &= \pi \int_{10-a}^{10} (100 - y^2) dy \\
 &= \pi \left[100y - \frac{y^3}{3} \right]_{10-a}^{10} \\
 &= \pi \left[\left(1000 - \frac{1000}{3} \right) - \left(100(10-a) - \frac{(10-a)^3}{3} \right) \right] \\
 &= \pi \left[1000 - \frac{1000}{3} - \left(1000 - 100a - \frac{1000 - 300a + 30a^2 - a^3}{3} \right) \right] \\
 &= \pi \left[-\frac{1000}{3} + 100a + \frac{1000}{3} - 100a + 10a^2 - \frac{a^3}{3} \right] \\
 &= \pi \left[10a^2 - \frac{a^3}{3} \right] \\
 &= \frac{\pi}{3} a^2 (30 - a)
 \end{aligned}$$

$$b) \text{ Find } \frac{dx}{dt} \text{ when } x=2 \quad \frac{dV}{dt} = 2\pi$$

$$\begin{aligned}
 V &= \frac{\pi}{3} x^2 (30 - x) \\
 &= \frac{\pi}{3} x^2 \cdot 30 - \frac{\pi}{3} x^3 \\
 &= 10\pi x^2 - \frac{\pi}{3} x^3
 \end{aligned}$$

$$\frac{dV}{dx} = 20\pi x - \pi x^2 \quad \text{At } x=2, \quad \frac{dV}{dx} = 40\pi - 4\pi = 36\pi$$

$$\begin{aligned}
 \frac{dx}{dt} &= \frac{dx}{dV} \times \frac{dV}{dt} \\
 &= \frac{1}{36\pi} \times 2\pi \\
 &= \frac{1}{18} \text{ cm/min}
 \end{aligned}$$

$$\begin{aligned}
 c) i) R &= \sqrt{(\sqrt{3})^2 + (1)^2} \\
 &= 2 \quad \tan \alpha = \frac{1}{\sqrt{3}} \\
 &\quad \alpha = \frac{\pi}{6}
 \end{aligned}$$

$$\therefore \sqrt{3} \sin \frac{t}{2} + \cos \frac{t}{2} = 2 \sin \left(\frac{t}{2} + \frac{\pi}{6} \right)$$

$$ii) \quad x = 2 \sin \left(\frac{t}{2} + \frac{\pi}{6} \right)$$

$$\begin{aligned}
 v &= \frac{1}{2} \cdot 2 \cos \left(\frac{t}{2} + \frac{\pi}{6} \right) \\
 &= \cos \left(\frac{t}{2} + \frac{\pi}{6} \right)
 \end{aligned}$$

$$\begin{aligned}
 a &= -\frac{1}{2} \sin \left(\frac{t}{2} + \frac{\pi}{6} \right) \\
 &= -\frac{1}{4} \times 2 \sin \left(\frac{t}{2} + \frac{\pi}{6} \right)
 \end{aligned}$$

$$= -\frac{1}{4} x \quad \text{which is in SHM}$$

$$iii) \quad 0.5 = \cos \left(\frac{t}{2} + \frac{\pi}{6} \right)$$

$$\frac{t}{2} + \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \dots$$

$$\frac{t}{2} = \frac{\pi}{6}, \frac{3\pi}{2}, \frac{13\pi}{6}$$

$$t = \frac{\pi}{3}, 3\pi$$

Question 7

$$\begin{aligned}
 a) \quad \lim_{x \rightarrow 0} \frac{\sin 6x}{5x} &= \lim_{x \rightarrow 0} \frac{\sin 6x}{6x} \times \frac{6}{5} \\
 &= 1 \times \frac{6}{5} \\
 &= \frac{6}{5}
 \end{aligned}$$

$$b) i) \quad x = vt \cos \alpha$$

$$\therefore t = \frac{x}{v \cos \alpha}$$

$$\text{Sub into } y = vt \sin \alpha - \frac{1}{2} g t^2$$

$$y = v \left(\frac{x}{v \cos \alpha} \right) \sin \alpha - \frac{1}{2} g \left(\frac{x}{v \cos \alpha} \right)^2$$

$$= x \tan \alpha - \frac{g x^2}{2 v^2 \cos^2 \alpha}$$

$$= x \tan \alpha - \frac{g x^2}{2 v^2} \sec^2 \alpha$$

$$ii) \quad \text{Ball reaches maximum height when } \dot{y} = 0$$

$$0 = v \sin \alpha - g t$$

$$t = \frac{v \sin \alpha}{g}$$

$$\text{Substitute into } y = vt \sin \alpha - \frac{1}{2} g t^2$$

$$y = v \left(\frac{v \sin \alpha}{g} \right) \sin \alpha - \frac{1}{2} g \left(\frac{v^2 \sin^2 \alpha}{g^2} \right)$$

$$= \frac{v^2 \sin^2 \alpha}{g} - \frac{1}{2} \frac{v^2 \sin^2 \alpha}{g}$$

$$= \frac{v^2 \sin^2 \alpha}{2g}$$

$$iii) \quad g = \frac{v^2 \sin^2 \alpha}{2h}$$

$$\therefore y = x \tan \alpha - \left(\frac{v^2 \sin^2 \alpha}{2h} \right) x^2 \sec^2 \alpha$$

$$= x \tan \alpha - \frac{v^2 \sin^2 \alpha x^2 \sec^2 \alpha}{2h \cdot 2v^2}$$

$$= x \tan \alpha - \frac{x \sin^2 \alpha}{\cos^2 \alpha \cdot 4h}$$

$$= x \tan \alpha - \frac{x^2 \tan^2 \alpha}{4h}$$

$$= x \tan \alpha \left(1 - \frac{x \tan \alpha}{4h} \right)$$

14)  $x=10$
 $y=1.6$

$$y = x \tan \alpha \left(1 - \frac{x \tan \alpha}{4h} \right)$$

$$1.6 = 10 \tan 30^\circ \left(1 - \frac{10 \tan 30^\circ}{4h} \right)$$

$$1.6 = 10 \cdot \frac{1}{\sqrt{3}} \left(1 - \frac{\frac{10}{\sqrt{3}}}{4h} \right)$$

$$1.6 = \frac{10}{\sqrt{3}} \left(1 - \frac{10}{4h\sqrt{3}} \right)$$

$$1.6 \times \frac{\sqrt{3}}{10} - 1 = -\frac{10}{4h\sqrt{3}}$$

$$0.7228718 = \frac{10}{4h\sqrt{3}}$$

$$h = 1.9967$$

$$\approx 2m$$

15) $1 + x + x^2 + \dots + x^{2n-1}$ $S_n = \frac{a(r^n - 1)}{r - 1}$

$$S_{2n} = \frac{1(x^{2n} - 1)}{x - 1}$$

$$= \frac{x^{2n} - 1}{x - 1}$$

$$x^2 - x - 1 = 0$$

$$x^2 = x + 1$$

$$x^{2n} = (x + 1)^n$$

$$x^{2n} = \sum_{r=0}^n {}^n C_r x^r$$

$$x^{2n} - 1 = \sum_{r=1}^n {}^n C_r x^r$$

Also $x^2 - x - 1 = 0$

$$x^2 - x = 1$$

$$x(x - 1) = 1$$

$$x = \frac{1}{x - 1}$$

$$S_{2n} = \frac{x^{2n} - 1}{x - 1}$$

$$= x \cdot \sum_{r=1}^n {}^n C_r x^r$$

$$= \sum_{r=1}^n {}^n C_r x^{r+1}$$